IBM SKILLSBUILD INTERNSHIP PROJECT REPORT

Project Title: Data Analytics & AI Business Insights Report **Submitted By:** Shriram **Program:** IBM SkillsBuild Data Analytics & AI Internship **Organized By:** BharatCares in association with AICTE **Submission Date:** October 2026

1. EXECUTIVE SUMMARY

This project was undertaken by **Shriram** as part of the IBM SkillsBuild Internship Program to apply data analytics and artificial intelligence techniques to practical business datasets. The primary goal is to process raw data, clean inconsistencies, perform Exploratory Data Analysis (EDA), generate visualizations, and deliver data-backed actionable strategic insights.

2. PROBLEM STATEMENT & OBJECTIVES

Modern business ecosystems generate large volumes of complex data. Extracting meaningful trends requires structured analytics.

Core Objectives by Shriram:

1. Clean and preprocess raw dataset variables (handle missing data and outliers).
2. Uncover hidden consumer behavior patterns and statistical relationships using EDA.
3. Construct data visualizations to communicate analytical findings clearly.
4. Provide strategic recommendations to optimize business decision-making.

3. DATASET DESCRIPTION & PREPROCESSING

- **Dataset Used:** Customer Demographic & Purchasing Behavior Dataset.
- **Key Variables:** CustomerID, Age, Annual_Income_k$, Spending_Score, Category.
- **Data Processing Workflow:**
    - Checked and validated dataset structure and null values.
    - Computed summary statistics (Mean, Standard Deviation, Quartiles).
    - Feature correlation analysis across numeric attributes.

4. EXPLORATORY DATA ANALYSIS (EDA) & VISUAL HIGHLIGHTS

- **Income Distribution:** Analysis revealed normal distribution spread across income categories.
- **Spending Behavior:** Identified key clusters linking customer income levels to overall spending score.
- **Category Trends:** Evaluated average customer spending across categories (Electronics, Clothing, Home, Books).

5. CONCLUSION & RECOMMENDATIONS

- **Conclusion:** The analytical pipeline built by Shriram successfully converted raw metrics into operational intelligence.
- **Key Recommendations:**
    1. Target marketing campaigns towards high-spending score customer segments.
    2. Implement data-driven stock optimization according to top-performing categories.
    3. Expand automated predictive modeling for future sales forecasting.

Report Prepared & Submitted By: Shriram **Internship Program:** IBM SkillsBuild Data Analytics & AI